📊 EXECUTIVE POWER BI DASHBOARD

Customer Shopping Behavior Dashboard

Interactive Power BI analysis of 3,900 consumer transactions across 50 US States



3,900

TOTAL CUSTOMERS

50 US States Covered

$59.76

AVG PURCHASE AMOUNT

$233,081 Total GMV

3.75 / 5.0

AVG REVIEW RATING

37 Nulls Median-Imputed

From Raw Data to Strategic Growth

How modern data engineering & analytics uncovers consumer purchase drivers

THE CORE BUSINESS QUESTION

"How can the company leverage consumer shopping data to identify demographic trends, optimize discounting strategies, and maximize customer loyalty?"



Python & Pandas

- Imputed 37 missing ratings via category medians
- Quartile-based age binning (`age_group`)
- Dropped redundant promo code columns



PostgreSQL (SQL)

- 10 Analytical Business Queries
- Window functions (`ROW_NUMBER`)
- Customer segmentation (New / Returning / Loyal)



Power BI

- Interactive multi-dimensional slicers
- Category & Age revenue distribution
- Clean KPI visual telemetry



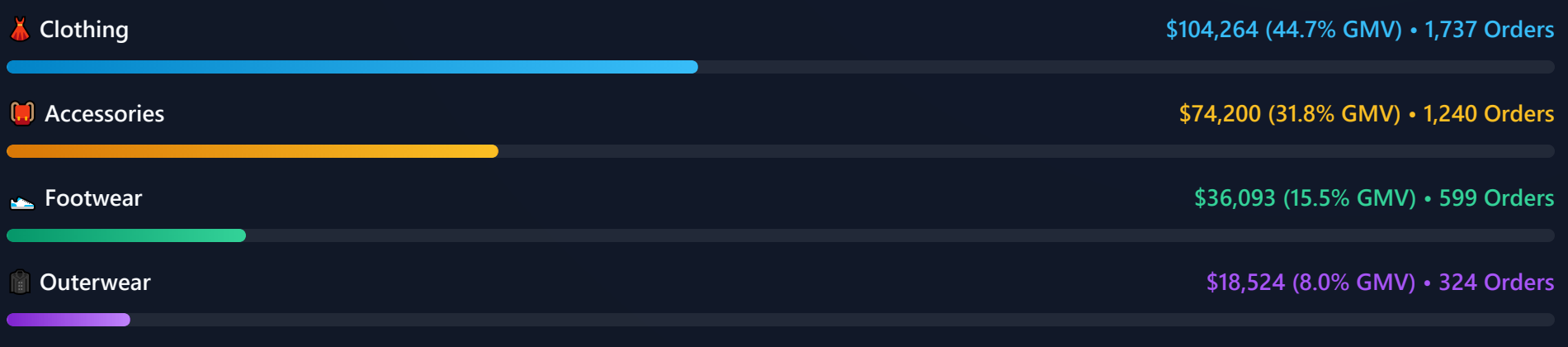
Dataset Scope: 3,900 customer records, 18 distinct features spanning transaction values, shipping preferences, subscription status, and purchase frequencies.

$233.1K Revenue Across 3,900 Purchases

Clothing and Accessories generate over 76% of total company revenue

$233.1K	3,900	$59.76	50
TOTAL GMV	TRANSACTIONS	AVG ORDER VALUE	US STATES

Category Revenue & Volume Breakdown



🔑 **Insight:** Clothing & Accessories represent **76.5% of total sales**. Footwear has the highest average spend per item ($60.26), signaling premium upsell potential.

The Subscription Paradox: A $100K Opportunity

72% of repeat buyers haven't subscribed — high loyalty without enrollment

NON-SUBSCRIBERS (73%) — 2,847 Shoppers

$170.4K

TOTAL REVENUE GENERATED

$59.87 Avg Spend / Order

SUBSCRIBERS (27%) — 1,053 Shoppers

$62.6K

TOTAL REVENUE GENERATED

$59.49 Avg Spend / Order

The Eye-Opening SQL Discovery (Q9):

Out of **3,476 repeat buyers** (>5 previous purchases), **2,518 customers (72.4%)** are **NOT subscribed**! Only 958 repeat customers hold an active subscription.

❌ Unsubscribed Repeat Shoppers: **2,518 (72.4%)** ✅ Subscribed Repeat Shoppers: **958 (27.6%)**

🎯 **Strategic Takeaway:** Customers are inherently loyal to the products, but the current subscription program offers zero clear incentive. Adding subscriber-exclusive discounts or free expedited shipping can unlock recurring revenue.

Gender Dynamics: Volume vs Basket Value

Male shoppers dominate transaction volume, but women deliver equal basket size

👦 Male Shoppers

68.0% OF ORDERS

$157,890

TOTAL REVENUE (67.7%)

Total Orders:	**2,652 customers**
Avg Basket Value:	**$59.54**

👩 Female Shoppers

32.0% OF ORDERS

$75,191

TOTAL REVENUE (32.3%)

Total Orders:	**1,248 customers**
Avg Basket Value:	**$60.25 (+$0.71)**

Strategic Marketing Implications

Protect Men's Footprint: Cross-sell accessories and footwear to the established 2.6K male shopper base to push AOV above $65.

Expand Women's Catalog: Female buyers have higher AOV ($60.25). Scaling acquisition campaigns can expand the 32% volume share.

Discounting: Margin Drain or Basket Builder?

839 high spenders use discounts; but 5 product lines have 47-50% discount dependency

SQL FINDING (Q2): THE "SMART SHOPPERS"

839 Customers Used a Discount & Still Outspent Average

Discounts drove basket enlargement rather than just eroding unit profitability.

839

HIGH-SPEND SHOPPERS

Top 5 Products with Highest Discount Application Rate (Q6)

Item Name	Category	Total Sold	Discounted Units	Discount Rate
🧢 Hat	Accessories	154	77	50.00%
👟 Sneakers	Footwear	145	72	49.66%
🧥 Coat	Outerwear	161	79	49.07%
🧶 Sweater	Clothing	164	79	48.17%
👖 Pants	Clothing	171	81	47.37%

⚠️ **Prescription:** Half of all sneaker and coat purchases rely on discounts. Transition from blanket promotions to **spend-threshold discounts** (e.g., "$15 off on $75+").

Product Stars & The Express Shipping Premium

Highest-rated product lines & how shipping speed correlates with basket value

★ Top 5 Rated Products (Q3)

🧤 Gloves	3.86 ★
👡 Sandals	3.84 ★
👢 Boots	3.82 ★
🧢 Hat	3.80 ★
👗 Skirt	3.79 ★

📦 Shipping Speed vs Basket Size (Q4)

⚡ 2-Day Shipping	$60.73 AOV
🚀 Express	$60.48 AOV
🎁 Free Shipping	$60.41 AOV
🏪 Store Pickup	$59.89 AOV
📦 Standard Shipping	$58.46 AOV

💡 **Key Takeaway:** Express and 2-Day shoppers spend **nearly 4% more per transaction** than Standard shipping customers. Offering complimentary 2-day delivery on carts over $70 creates an immediate incentive for basket enlargement.

The 80% Loyalty Engine & Age Distribution

3,116 customers are highly loyal repeat buyers; balanced demand across all cohorts

👑

3,116

LOYAL SEGMENT (79.9%)

>10 Previous Purchases

🔄

701

RETURNING (18.0%)

2–10 Previous Purchases

🌱

83

NEW SHOPPERS (2.1%)

1st Lifetime Purchase

Revenue Contribution by Age Cohort (Q10)

Young Adult (18-31)	Middle-Aged (45-54)	Adult (32-44)	Senior (55-70)
$62,143	**$59,197**	**$55,978**	**$55,763**
1,028 Orders (26.7%)	986 Orders (25.4%)	942 Orders (24.0%)	944 Orders (23.9%)

📌 **Takeaway:** Revenue is remarkably distributed across all age demographics (~24-27% each). Young adults lead total GMV, while seniors demonstrate identical spending capacity.

4 Actionable Executive Recommendations

Targeted business initiatives to capture recurring revenue and protect margins

1. Monetize Repeat Shoppers

72.4% of repeat buyers (>5 orders) are not subscribed. Launch a tiered membership featuring free express delivery and members-only flash drops to capture predictable recurring cashflow.

2. Spend-Threshold Discounts

50% of footwear and outerwear purchases rely on discounts. Phase out blanket flat discounts; replace with minimum order thresholds ($75+) to protect unit margins while boosting cart size.

3. Cross-Category Bundling

Pair high-volume Clothing staples (Blouses, Shirts) with high-margin Accessories (Jewelry, Sunglasses) at checkout. Accessories boast high attach rates and expand overall basket margin.

4. Incentivize Expedited Shipping

Express shoppers spend 4% more per order ($60.73 vs $58.46). Subsidize 2-Day Shipping on orders exceeding $70 to entice standard shipping customers into higher spend brackets.

💼 **Projected Outcome:** 15-20% boost in subscription enrollment, 8-12% expansion in AOV, and margin preservation on core seasonal lines.

End-to-End Retail Data Architecture

Tools used, code repository, and let's start the discussion

Technology Stack Summary

🐍 Python (Pandas)
Missing rating imputation, schema standardization, feature binning.

🐬 PostgreSQL
10 Complex business queries, window functions, RFM customer segmentation.

📊 Power BI
Interactive dashboard, KPI visual telemetry, dynamic cross-filtering.

LET'S CONNECT & DISCUSS

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💬 Question for the community:

"Which metric surprised you most? The 72% unsubscribed repeat buyers or the discount dependency on footwear? Share your thoughts below!"

🔗 github.com/DheerajChavhan/Data-Analysis